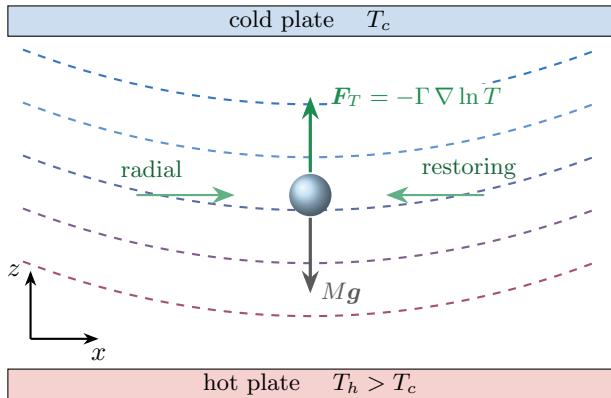


Harmonic thermophoretic trap



isotherms

$$\nabla \ln T = \mathbf{A} \mathbf{x} + \mathbf{G}_0$$

$\mathbf{A} = \text{diag}(\alpha_r, \alpha_r, \alpha_z)$: trap curvature

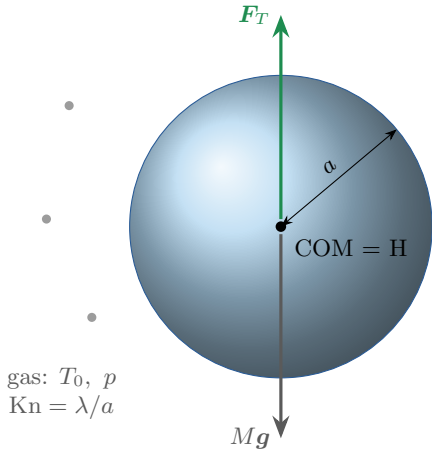
$\mathbf{G}_0 = -g_0 \hat{e}_z$: background gradient

Balanced: $g_0 = Mg [(\mathbf{R}_0 \Gamma \mathbf{R}_0^T)^{-1}]_{zz}$

Trap frequencies: $\alpha_i = M(2\pi f_i)^2 / \Gamma_{ii}$

Required gradient: $|\nabla T| = g_0 T_0$

1. Sphere



Parameters: a, ρ_p, k_p ; gas T_0, p

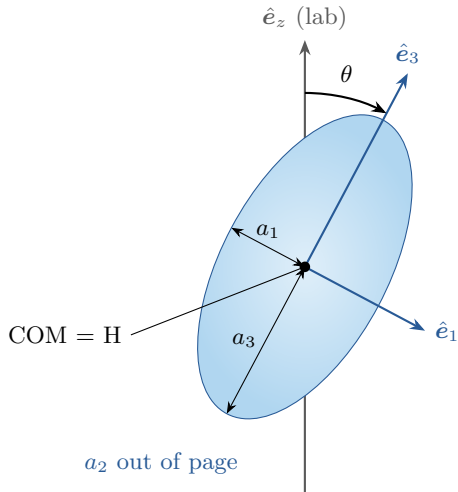
$$M = \frac{4}{3}\pi a^3 \rho_p, \quad I = \frac{2}{5}Ma^2 \mathbf{1}$$

$$\mathbf{K} = \frac{6\pi\mu a}{C_c(\text{Kn})} \mathbf{1}, \quad \Omega = \frac{8\pi\mu a^3}{1 + 3\ell_r/a} \mathbf{1}$$

$$\mathbf{B} = 0, \quad \Gamma = \gamma(a, \text{Kn}) \mathbf{1}, \quad \Lambda = 0$$

γ : APL2017 fit, Waldmann, or Talbot

2. Uniform ellipsoid (tilted by θ)



Parameters: a_1, a_2, a_3, ρ_p ; gas T_0, p

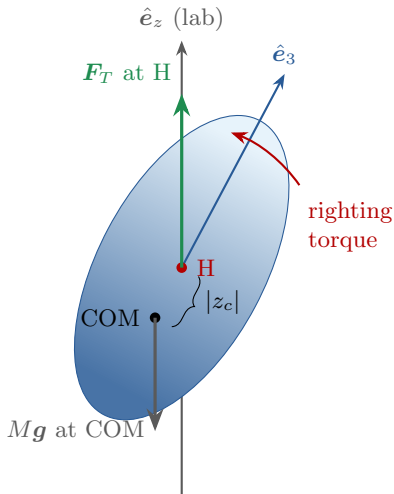
$$\mathbf{K} = \text{diag}(K_1, K_2, K_3), \quad K_i = \frac{16\pi\mu}{\chi + a_i^2\alpha_i}$$

$$\Omega_{11} = \frac{16\pi\mu(a_2^2 + a_3^2)}{3(a_2^2\alpha_2 + a_3^2\alpha_3)}, \text{ etc.}$$

$\Gamma = v_T \mathbf{K}$ (isotropic thermophoretic velocity), $\Lambda = 0$

Tilted: $\mathbf{F}_T \nparallel \nabla \ln T \Rightarrow$ sideways drift

3. Graded ellipsoid, $\rho = \rho_p(1 + \varepsilon z)$, $\varepsilon < 0$



Parameters: $a_1, a_2, a_3, \rho_p, \varepsilon$ (with $|\varepsilon|a_3 < 1$)

M unchanged; $z_c = \varepsilon a_3^2/5$ (COM shift)

$I_{\text{COM}} = I_{\text{unif}} - M z_c^2 \text{diag}(1, 1, 0)$

Tensors defined at H, shifted to COM via $\mathbf{r}_H = -z_c \hat{e}_3$:

$\mathcal{R}_{\text{COM}} = \mathbf{P}^\top \mathcal{R}_H \mathbf{P}$, $\mathcal{T}_{\text{COM}} = \mathbf{P}^\top \mathcal{T}_H$

Stiffness $\kappa = Mg|z_c| \Gamma_{11}/\Gamma_{33}$, $\omega_p^2 \simeq \kappa/I_\perp$